retrospect, his reticence was due largely to our good-natured and long-standing intellectual joust in which we each play devil's advocate to the other's ideas. Within days, he came around and we turned our full attention to flops.

By then, Morrison had also arrived, and the three of us met in the Institute's tea-room to formulate a strategy. We agreed that the central goal was to determine whether the evolution from Figure 11.3(a) to Figure 11.4(d) can actually occur in our universe. But a direct attack on the question was forbidding, because the equations describing this evolution are extremely difficult, especially when the spatial tear occurs. Instead, we chose to rephrase the issue using the mirror description, hoping that the equations involved might be more manageable. This is schematically illustrated in Figure 11.5, in which the top row is the original evolution from Figure 11.3(a) to Figure 11.4(d), and the bottom row is the same evolution from the perspective of the mirror Calabi-Yau shapes. As a number of us had already realized, it turns out that in the mirror rephrasing it appears that string physics is perfectly well behaved and encounters no catastrophes. As you can see, there does not seem to he any pinching or tearing in the bottom row in Figure 11.5. However, the real question this observation raised for us was this: Were we pushing mirror symmetry beyond the bounds of its applicability? Although the upper and lower Calabi-Yau shapes drawn on the far left-hand side of Figure 11.5 yield identical physics, is it true that at every step in the evolution to the right-hand side of Figure 11.5—necessarily passing through the pinch-tear-repair stage in the middle—the physical properties of the original and mirror perspective are identical?

Although we had solid reason to believe that the powerful mirror relationship holds for the shape progression leading to the tear in the upper Calabi-Yau shape in Figure 11.5, we realized that no one knew whether the upper and lower Calabi-Yau shapes in Figure 11.5 continue to be mirrors after the tear has occurred. This is a crucial question, because if they are, then the absence of a catastrophe in the mirror perspective would mean an absence in the original, and we would have demonstrated that space can tear in string theory. We realized that this question could be reduced to a calculation: Extract the physical properties of the universe for the upper Calabi-Yau shape